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Education hubs in a globalized world: The emergence of China

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ABSTRACT

Education hubs represent the third generation of crossborder education activities, which emerge onto the landscape of our more globalized world. As a new phenomenon in higher education, education hubs are appearing in different locations around the globe. In the literature on education hubs, most research studies have focused on Southeast Asia and Middle East. To address this gap in the literature, we seek to explore the emergence of China as an international education hub, especially with respect to its impact on development. Against the backdrop of China's national strategy to develop its international competitiveness in education, we conduct case studies of three education hubs in China: (i) the International Collaborative Education Exemplary Zone by the International Campus of Zhejiang University (iZJU) in the Yangtze River Delta; (ii) the International Higher Education Demonstration Zone in the Guangdong-Hong Kong-Macau Greater Bay Area; and (iii) the International Education Innovation Pilot Zone in the Hainan Free Trade Port. Overall, our paper draws implication on how education hubs developed through university-industry-government collaboration can help the region to shift to a knowledge and service-based economy, and gain a competitive edge within the country and beyond.

1. Introduction

With the concept of 'universe' in the naming of 'universities', higher education has had an international dimension for centuries. Internationalization is changing the world of higher education, and globalization is changing the world of internationalization. In higher education, globalization and internationalization are different but related. Globalization is defined as the broad economic, scientific, and technological trends that directly affect higher education (Altbach, 2004, 2007). Internationalization is defined as the process of integrating an international, intercultural, or global dimension into the purpose, functions, and delivery of higher education (Knight, 2004: p.11). Pushed and pulled along by the forces of globalization (Altbach, 2009), internationalization includes the policies and practices that are undertaken by higher education systems and institutions to cope with globalization (Altbach and Knight, 2007). Globalization may be unalterable, but internationalization can involve many choices. Both as an actor and as a reactor to globalization, internationalization has been called upon to help contribute to tackle the complex global challenges faced by global society today (de Wit and Altbach, 2021: p.29).

Internationalization has not only transformed higher education, but it has undergone fundamental changes itself. There are two pillars in the internationalization process of higher education: (1) 'Internationalization at Home' (IaH, Nilsson, 2003); and (2) 'crossborder education', that is, internationalization abroad. These two pillars are closely linked and interdependent. The IaH concept is defined as the purposeful integration of international and intercultural dimensions into the formal and informal curriculum for all students in domestic learning environments (Beelen et al., 2015: p.69). Within the context of crossborder education, there are three generations of development (Knight, 2011: p.223). The *first* generation is academic mobility, which is about the movement of students and scholars around the world. The *second* generation is program and provider mobility, which is about the movement of degree programs and higher education institutions between countries. Education hubs build on the first and second generations of crossborder education activities, and constitute the *third* generation of crossborder education initiatives, which demonstrate the responsive nature of internationalization to the complex realities of our more globalized world.

In the literature on higher education, an education hub is defined as a planned effort to build a critical mass of local and international actors strategically engaged in education, training, knowledge production, and innovation initiatives (Knight, 2011: p.233). As a new phenomenon in the landscape of higher education, education hubs are appearing in

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Table 1
literature survey: education hubs – by region & economy.

region	economy	literature	type[a]
Asia	Singapore	Olds (2007); Mok (2008); Knight (2011); Mok (2011); Mok and Yu (2011); Sidhu, Ho, and Yeoh (2013); Lee (2014, 2015); Soong (2020).	(3)
	Malaysia	Knight (2011); Knight and Morshidi (2011); Mok (2011); Mok and Yu (2011); Aziz and Abdullah (2013); Cheng, Mahmood, and Yeap (2013); Lee (2014, 2015).	(1)
	Hong Kong	Cheng, Cheung, and Yeun (2011); Knight (2011); Lai and Maclean (2011); Lo (2013); Mok and Yu (2011); Mok and Bodycott (2013); Lee (2014, 2015); Mok (2016).	(3)
	Phillipines	Ortiga (2018).	(1)
	Thailand	Pongsin, Lawthong, Fry, Ransom, Kim, and My (2023).	(1)
	South Korea	Dou and Knight (2013); Jon, Lee and Byun (2014).	(1)
	Sri Lanka	Dou and Knight (2013).	(1)
Middle East	UAE	Knight (2011); Fox et al. (2013).	(2)
	Qatar	Knight (2011); Ibnouf, Dou, and Knight (2013).	(2)
	Bahrain	Knight (2011); Dou and Knight (2013).	(1)
Africa	Botswana	John, Wilmoth, and Mokopakgosi (2013).	(1)
	Mauritius	Dou and Knight (2013); Knight and Motala-Timol (2022).	(1)

[a] In Table 1, 'type' refers to the type of education hubs, where (1) = student hub, (2) = talent hub, and (3) = knowledge-innovation hub.

different regions of the world. Based on the rationales of the activities, there are three different but linked models of education hubs, which include: student hub, talent hub, and knowledge-innovation hub (Knight, 2011: p.234; Knight and Lee, 2013: p.30). A *student hub* has the key activity of education and training of local and international students, and it has the primary objectives to modernize and internationalize domestic higher education institutions, and to increase competitiveness in the regional higher education sector and beyond. A *talent hub* has the overarching goal to develop a skilled labor force. A *knowledge-innovation hub* broadens its mandate beyond education and training to include the production and distribution of knowledge and innovation, with the primary objectives to help build a knowledge and service-based economy, and increase regional or global economic competitiveness.

In the literature on education hubs, most research studies have focused on Southeast Asia and Middle East (Table 1). China is commonly perceived as a major sending country of international students, but it is often overlooked as an important receiving country of international students (Wei, 2013; Ma, 2017; Wen, Hu, and Hao, 2018; Wen and Hu, 2019; Tian and Liu, 2021),[1] and as a regional education hub (Mok and Yu, 2011; Wen and Hu, 2019). To address this gap in the literature, this paper seeks to explore the emergence of China as an international education hub, especially with respect to the impact on development.

Against the backdrop of China's national strategy to develop the country's international competitiveness in education, China has rapidly become one of the most popular destinations for international students.[2] In 2019, China emerged as the No. 1 study abroad destination in Asia[3]. In 2020, China rose to be the No. 4 study abroad destination in the world. With the implementation of the 'China's Education Modernization 2035 Plan',[4] China's education sector is entering into a new era,[5] which would act as an incentive for more Chinese students to remain in China for study, that is, 'Internationalization at Home' (Qu and Dai, 2024), and further enhance China's reputation as a destination for international students. More recently at the 5th group study session of the Political Bureau of the CPC Central Committee in May 2023, China's President Xi Jinping called for accelerated efforts to build a leading country in education,[6] as part of the Chinese path to modernization[7]; and to make China an important education center with strong influence in the world, along with the brand of 'Study in China'. Therefore, it is of both academic interest and practical relevance to examine the case of China as an emerging international education hub.

The remainder of this paper proceeds as follows. In *Section* 2, we review the development and strategies in the internationalization and globalization of the higher education sector in China. In *Section* 3, we conduct case studies of three education hubs in China: (i) the International Collaborative Education Exemplary Zone by the International Campus of Zhejiang University (iZJU) in the Yangtze River Delta; (ii) the International Higher Education Demonstration Zone in the Guangdong-Hong Kong-Macau Greater Bay Area; and (iii) the International Education Innovation Pilot Zone in the Hainan Free Trade Port. *Section* 4 concludes.

2. The internationalization and globalization of higher education in China

In the late 1970s, China set the goal of the 'Four Modernizations' to strengthen the sectors of agriculture, industry, science and technology, and defense.[8] An important aspect of modernization was education. Following the rapid expansion of higher education in 1998,[9] the Chinese higher education system has been going through restructuring along the lines of marketization, privatization, and decentralization,[10] and it has been transformed through the processes of massification,

[1] "Is China both a source and hub for international students?", China Power Project, Center for Strategic & International Studies (CSIS), Washington DC, 2020. See https://chinapower.csis.org/china-international-students/

[2] James, F. (2022), "Three leading factors in China' booming popularity as an international study destination", QS. See https://www.qs.com/three-leading-factors-in-chinas-booming-popularity-as-an-international-study-destination/

[3] "China emerges as the No. 1 study abroad destination in Asia", 2019. See https://www.studyinternational.com/news/china-emerges-as-the-no-1-study-abroad-destination-in-asia/

[4] "中共中央 国务院印发《中国教育现代化2035》", Beijing, February 2019. See https://www.gov.cn/xinwen/2019–02/23/content_5367987.htm

[5] "China's education is entering into a new era", Ministry of Education (MoE), Beijing, March 2019. See http://en.moe.gov.cn/features/Specials_twoSessions/twosessions_Opinions/201903/t20190329_375938.html

[6] "习近平在中共中央政治局第五次集体学习时强调加快建设教育强国为中华民族伟大复兴提供有力支撑", Beijing, May 29, 2023. See https://www.gov.cn/yaowen/liebiao/202305/content_6883632.htm

[7] "Xi calls for accelerated efforts to build leading country in education", *China Daily*, May 30, 2023. See https://www.chinadaily.com.cn/a/202305/30/WS6475f508a310b6054fad5d76.html

[8] See Yang (2014: p.151); Yang (2016: p.35); Mok and Marginson (2021: p.2).

[9] See Huang (2015: p.204); Mok and Marginson (2021: p.1).

[10] See Mok and Xu (2008: p.393); Mok and Yu (2011); Ennew and Yang (2009: p.26); Jiang and ke (2021: p.430); Mok (2021a: p.414); Mok (2021b: p.2); Mok, Ke, and Tian (2022: p.4).

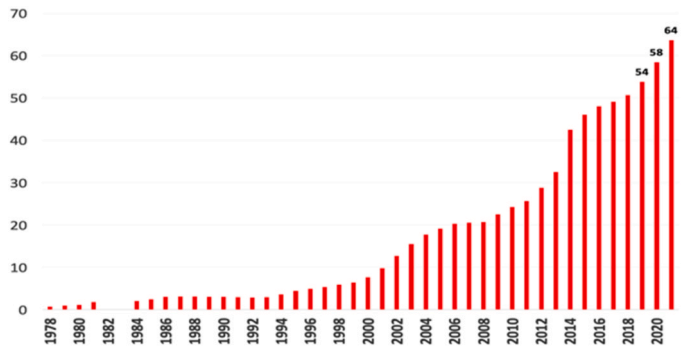


Fig. 1. School enrollment, tertiary (% gross) – China. *Data Source*: UNESCO Institute for Statistics (UIS), World Bank.

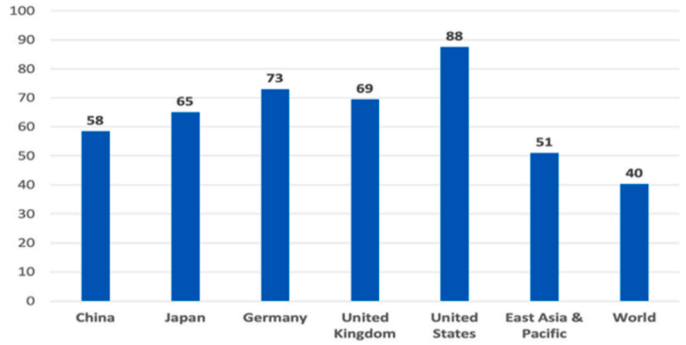


Fig. 2. School enrollment, tertiary (% gross): 2020. *Data Source*: UNESCO Institute for Statistics (UIS), World Bank.

diversification, and internationalization (Mok and Marginson, 2021) from an elite model to a mass one,[11] with a gross enrollment rate of 3% in 1990 to 8% in 2000 and 58% in 2020 (Fig. 1), which is still lower than that of advanced economies, but higher than East Asia and Pacific, and the world (Fig. 2). China has built the world's largest higher education system (Postiglione, 2015; Yang, 2018). Since the call for the massification of higher education, and the quest for world-class university status,[12] Chinese universities, especially top-tier ones, have developed strategies to strengthen fundamental sciences, encourage interdisciplinary research, improve faculty quality, serve the needs of the nation, and promote internationalization (Huang, 2015: p.208).[13]

In the rapidly changing globalized world, there are three major approaches taken in the internationalization of the higher education sector in China, which include: study abroad, curriculum internationalization, and transnational higher education (Yang, 2014; Yang, 2016). *Studying abroad* includes dispatching Chinese students and staffs abroad for advanced studies or research, and attracting foreign students (Wu, 2019).[14] *Curriculum internationalization* includes integrating an international dimension into university teaching and learning, such as introducing foreign textbooks and references, developing bilingual courses. *Transnational higher education*[15] includes providing transnational programs in cooperation with foreign institutional partners, such as Sino-foreign joint programs and Sino-foreign joint institutes (Hu and Willis, 2017). Overall, these three approaches of internationalization have drastically improved the Chinese higher education sector in both quality and quantity, which in turn have led more Chinese students to remain in China for study, that is, 'Internationalization at Home' (Qu and Dai, 2024), and more international students to come to China for study.

Drawing on the range of reform since China's opening up in 1978, the Chinese State Council published the 'China's Education Modernization 2035 Plan' (Zhu, 2019), to drive continued reform in and advancement of China's education sector. The 2035 Plan is aimed at substantially modernizing China's education system by 2035, which is expected to act as an incentive for more Chinese students to remain in China for study, and further enhance China's reputation as a destination for international students. In the 2035 Plan, some regions are prioritized, mainly eastern and coastal regions that have innovative development agendas, such as the Yangtze River Delta, Guangdong-Hong Kong-Macau Greater Bay Area (GBA), and the Hainan Free Trade Port (FTP).

More recently at the 5th group study session of the Political Bureau of the CPC Central Committee in May 2023, China's President Xi Jinping called for accelerated efforts to build a leading country in education, to improve the strategy of opening up education to the outside world, to coordinate the two major tasks of 'bringing in' and 'going out', to effectively utilize world-class education resources and innovative elements, and to make China an important education center with strong

[11] See Huang (2003); Huang (2015: p.204); Chen (2004); Yang (2004); Yang (2018: p.2); Gao (2018); Mok and Xu (2008: p.394); Ennew and Yang (2009: p.22); Postiglione (2015: p.237); Jiang and Ke (2021); Mok (2021a: p.399); Mok (2021b: p.2); Mok and Kang (2021); Mok and Marginson (2021: p.5); Mok, Ke, and Tian (2022: p.3).

[12] See Mok (2007: p.445); Mok and Xu (2008: p.394); Ngok and Guo (2008); Huang (2015); Postiglione (2015: p.237); Liu and Metcalfe (2016: p.400); Wu (2019: p.85); Li and Xue (2021); Mok (2021a: p.403); Mok (2021b: p.4); Mok and Kang (2021); Mok and Marginson (2021: p.2); Guo, Guo, Yochim, and Liu (2022).

[13] See Ngok and Guo (2008: p.550); Huang (2015: p.208); Guo, Guo, Yochim, and Liu (2022).

[14] There are some foreign national leaders who have also studied in China, including the Vietnamese Deputy Premier, Thai Princess, Kazakhstan Prime Minister, Ethiopian President (Yang, 2019: p.3).

[15] See Mok and Xu (2008); Yang (2008); Mok and Marginson (2021: p.2); Mok and Han (2016, 2017); Mok (2021a).

influence in the world, along with the brand of 'Study in China'.

3. Case studies: Education hubs in China

As part of China's regional development strategy, coordinated regional development is an objective which involves implementing both a strategy for overall coordination and a strategy that is focused on developing the main functional areas (Du, 2014). There are four coordinated development areas in China, including the Beijing-Tianjin-Hebei Economic Circle (Yu, 2015), the Yangtze River Delta Agglomeration,the Chengdu-Chongqing Economic Circle, and the Yangtze River Middle Reaches Urban Agglomeration (Fig. 3). The regional integration of these four coordinated development areas are mutually reinforcing and jointly promoting high-quality growth in China.[16] Based on their development experiences, the development of the Guangdong-Hong Kong-Macao Greater Bay Area (GBA) is also accorded the status of key strategic planning in China's development blueprint.

In the literature, higher education plays a critical role in regional development (Harrison and Turok, 2017), with different drivers and dynamics at work, such as the 'triple helix' model of university-industry-government relations (Etzkowitz and Leydesdorff, 2000); the intersection of the learning region and local and regional economic development (Shaw and Allison, 1999; Maskell et al., 2003); regional innovation systems (Benneworth, Pinheiro, and Karlsen, 2017). In practice, the Chinese government has also engaged higher education institutions in regional development (Wang and Vallance, 2015; Xie, Huang, and Jung, 2022).

In this section, we conduct case studies of three education hubs in China, which include (i) the International Collaborative Education Exemplary Zone by the International Campus of Zhejiang University (iZJU). It is based in the Yangtze River Delta, which is one of China's four coordinated development areas and is also one prioritized region in China's 2035 Plan; (ii) the International Higher Education Demonstration Zone in the Guangdong-Hong Kong-Macao Greater Bay Area; and (iii) the International Education Innovation Pilot Zone in the Hainan Free Trade Port.

3.1. *The iZJU International Collaborative Education Exemplary Zone (iZJU国际合作教育样板区)*

Located in the Yangtze River Delta of China, Zhejiang University (ZJU) is among the top 50 universities in the world,[17] and it is a member of the C9 League (Huang, 2015: p.207). At the forefront of higher education, ZJU has been striving to become a pioneer of reform and an exemplar of development,[18] and it has set its goal of becoming a world-class university (Xu, 2008)[19] with Chinese characteristics (Li and Xue, 2021: p.14). To accelerate its efforts, ZJU began to build its international campus in 2013, which came into use in 2016, as a base for international collaboration.

The International Campus of Zhejiang University (iZJU) is a pioneer of 'Internationalization at Home' (Qu and He, 2022; Qu and Dai, 2024) and an exemplar of international collaborative education in China. As a pioneer of 'Internationalization at Home', iZJU is China's first campus for transnational higher education,[20] with the unique characteristic of '1 +X′, that is, iZJU can establish more than one Sino-foreign joint institutes on one campus.[21] There are currently two Sino-foreign joint institutes at iZJU (Table 2): Zhejiang University – University of Edinburgh Institute (the ZJU-UoE Institute, or ZJE), and Zhejiang University – University of Illinois at Urbana-Champaign Institute (the ZJU-UIUC Institute, or ZJUI). [22] By partnering with world-class universities, iZJU seeks to combine the best practices of Eastern and Western education, so as to train globally minded and high-calibre innovators, and cultivate future leaders with a global vision.

As one of ZJU's eight campuses, iZJU has its own campus away from the main campus of ZJU in Hangzhou, but benefits from the knowledge spillovers among ZJU's network model of campuses (Fig. 4: left panel). As a part of ZJU, iZJU has an organizational structure with two tiers: at the campus and the institute levels. At the campus level, the iZJU leadership team acts on behalf of ZJU to oversee the operation of the campus and the two Sino-foreign joint institutes, with a coordinating committee and an academic affairs committee.

At the institute level, both of the two Sino-foreign joint institutes are operated by an iZJU-based academic and administrative team, a joint management committee (JMC), and an international advisory board (IAB), which together facilitate and coordinate the governance and management between ZJU and its foreign institutional partners, via a partnership model (Fig. 4: right panel).

As an exemplar of international collaborative education, the location of iZJU in the Yangtze River Delta is one of the prioritized regions in the 'China's Education Modernization 2035 Plan' in February 2019. The development of iZJU as an International Collaborative Education Exemplary Zone is a national strategy in the "Outline of the Yangtze River Delta Regional Integrated Development Plan" in December 2019.[23] Its construction plan was jointly issued by the National Development and Reform Commission (NDRC), the Ministry of Education (MoE), and the Ministry of Science and Technology of China in October 2022. Its implementation plan was jointly issued by the Development and Reform Commission, the Department of Education, and the Department of Science and Technology of Zhejiang Province in April 2023. According to the Construction Plan at the national level and the Implementation Plan at the provincial level (Table 3),[24] the development of iZJU as an International Collaborative Education Exemplary Zone has iZJU as the core, and links the development of the Haining city, which is a county-level city in Zhejiang Province.

As an exemplary zone for international collaborative education, iZJU aims to become a student hub, by expanding the scale, scope, and structure of international collaboration in education, with the feature of 'Internationalization at Home' for Chinese students (Qu and Dai, 2024), and the brand of 'Study in Haining' for international students. Moreover, iZJU aims to become a knowledge-innovation hub and a talent hub, by cultivating a knowledge, innovation, and entrepreneurial ecosystem joint with the Juanhu International Science Park, and attracting world-class talent in education and science.

[16] "粤港澳、长三角、京津冀区域协调助推大国经济高质量发展", Liaison Office of the Central People's Government in the Hong Kong SAR, November 2021. See http://www.locpg.gov.cn/jsdt/2021–11/02/c_1211429388.htm

[17] According to the QS World Univesity Rankings 2024, Zhejiang University (ZJU) ranks No. 44 in the world. See https://www.topuniversities.com/university-rankings/world-university-rankings/2024?&page= 2

[18] Wu, Z. (2022a), " 'Seeking truth' as the foundation and 'pursuing innovation' as the soul", remarks at the gala celebrating the 125th anniversary of Zhejiang University, May 21. See https://www.zju.edu.cn/english/2022/0525/c19573a2554529/page.htm

[19] "A look into ZJU's plan to world-class university", ZJU website, 2017. See https://www.zju.edu.cn/english/2017/0629/c19573a811434/page.htm

[20] Wu, Z. (2022b), "Universities retain relevance in service to the common good", *University World News*. See https://www.universityworldnews.com/post.php?story= 20220903075811440

[21] "Zhejiang University to build Jiaxing campus role model of international co-op education", eZhejiang, 2020. See http://www.ezhejiang.gov.cn/2020–04/26/c_473772.htm

[22] O'Meara, S. (2018), "China's Zhejiang province is open for science business", *Nature*. See https://www.nature.com/articles/d41586–018–05265-z

[23] "中共中央 国务院印发《长江三角洲区域一体化发展规划纲要》", National Development and Reform Commission (NDRC), Beijing, December 2019. See https://www.ndrc.gov.cn/xwdt/ztzl/cjsjyth1/ghzc/202007/t20200728_1234708.html

[24] In China, policies are often promulgated nationally, but implemented locally (Sun, Lei, and Yin, 2018).

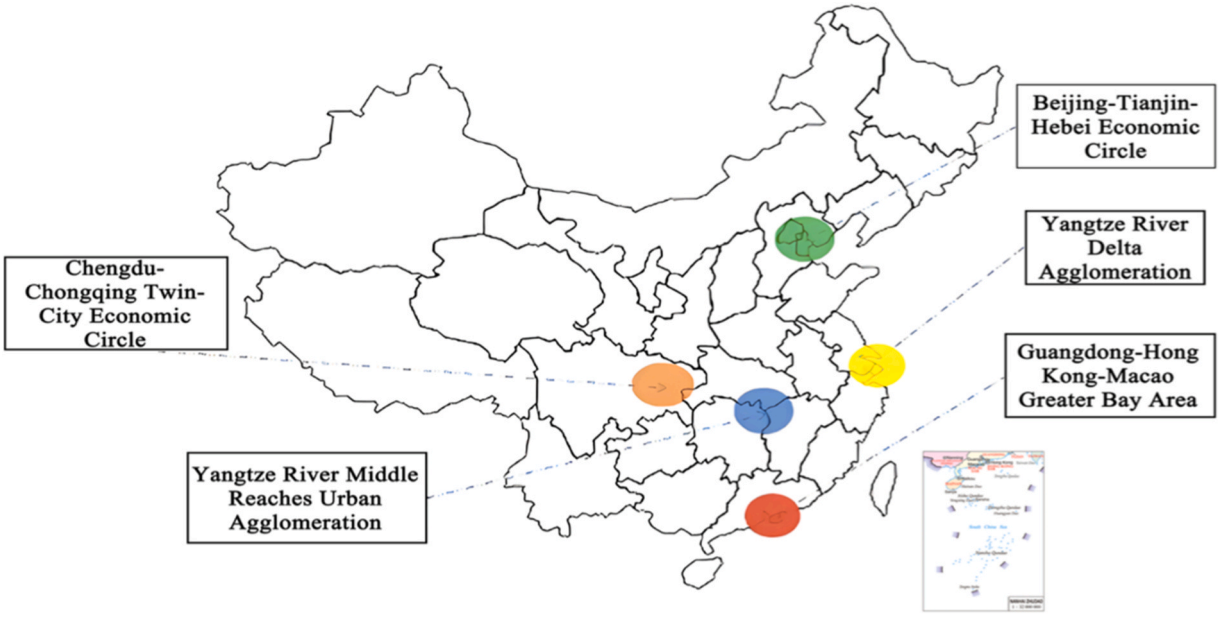


Fig. 3. China's four coordinated development areas and the Greater Bay Area (GBA).
Source: Xie, Liu, and McNay (2023: Figure 2).

Table 2
Sino-foreign joint institutes at iZJU.

Sino-foreign joint institute	Chinese university	foreign university	year
ZJU-UoE Institute (ZJE)	Zhejiang University (ZJU)	University of Edinburgh (UoE)	2016
ZJU-UIUC Institute (ZJUI)	Zhejiang University (ZJU)	University of Illinois at Urbana-Champaign (UIUC)	2016

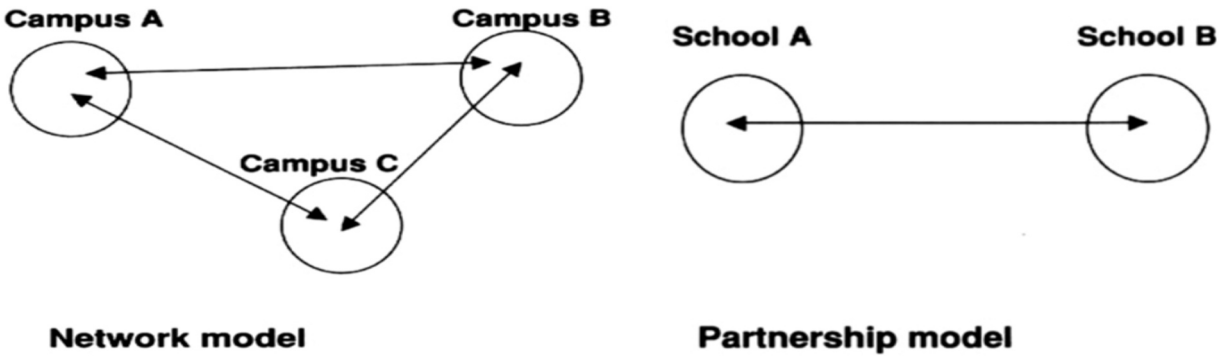


Fig. 4. The iZJU model at the campus level and at the institute level.

Overall, the regional development model of iZJU is like a single-core-driven model with iZJU as the core, which leverages the strength of ZJU's network model of campuses and iZJU's partnership model with world-class universities (Fig. 4), cultivates a knowledge, innovation, and entrepreneurial ecosystem with the Juanhu International Science Park that is conducive to attracting international talent, and integrates the development of iZJU and the Haining city. It sets a role model for the transformation of other county-level cities like Haining in China.

3.2. The GBA International Higher Education Demonstration Zone (粤港澳大湾区的国际高等教育示范区)

Drawing reference to other major bay area economies in Tokyo,[25]

Table 3
Main policy documents for iZJU as an International Collaborative Education Exemplary Zone.

	policy document
December 2019	Outline of the Yangtz River Delta Regional Integrated Development Plan. 《长江三角洲区域一体化发展规划纲要》
October 2022	Construction Plan of the International Campus of Zhejiang University (iZJU) as an International Collaborative Education Exemplary Zone. 《浙江大学国际联合学院(海宁)国际合作教育样板区建设方案》
April 2023	Implementation Plan of the Internatinoal Campus of Zhejiang University (iZJU) as an International Collaborative Education Exemplary Zone (2023–20225) by Zhejiang Province. 《浙江省推进浙江大学国际联合学院(海宁)国际合作教育样板区建设实施方案(2023–2025年)》

[25] See, for example, Deng (2022).

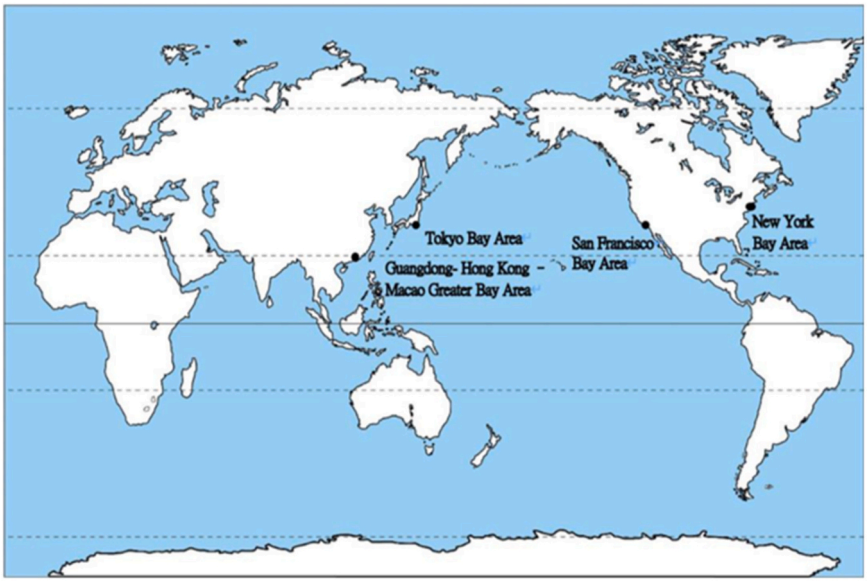


Fig. 5. The world's major bay areas. *Source*: Education Bureau, Hong Kong SAR.

San Francisco,[26] and New York,[27] China's National Development and Reform Commission (NDRC), and the government of Guangdong Province, Hong Kong SAR, Macau SAR signed the "Framework Agreement on Deepening Guangdong-Hong Kong-Macau Cooperation in the Development of the Greater Bay Area" in Hong Kong in July 2017,[28] which set out the goals and principles of cooperation in the development of the Greater Bay Area (GBA). This put the GBA as one of the four largest bay areas in the world (Fig. 5).

Among these major bay areas in the world, the Tokyo Bay Area is a metropolitan area that covers Tokyo and its seven surrounding prefectures. It has a single-core-driven model for regional development. The San Francisco Bay Area consists of many cities along the industrial belt of Silicon Valley and covers 9 counties bordering the San Francisco Bay. It has a world-class university cluster and a university-industry-research cooperation model. The New York Bay Area has a dual-core-driven model for regional development, with New York as its financial center and Boston as its educational center (Xie, Liu and McNay, 2023).

Located in the southern coastal area of China, the GBA is a megalopolis of a 11-city cluster, including nine cities in Guangdong Province, and the two special administrative regions of Hong Kong and Macau (Fig. 6). The GBA has a multi-core-driven model for regional development, with Guangzhou as an education center, Shenzhen as an innovation center, Hong Kong as a financial center, and Macau as a tourism center. Compared to the other three major bay areas, the GBA has the most extensive land area and the largest population, but it has the lowest per-capita GDP, and the lowest tertiary industry share (Table 4).

In 2019, the Chinese central government promulgated the "Outline Development Plan for the Guangdong-Hong Kong-Macau Greater Bay Area", which signified a new milestone in the development of the GBA. As a national strategy, the objectives of the GBA are to further deepen cooperation amongst Guangdong, Hong Kong SAR, and Macau SAR, fully leverage the comparative advantage of these three places, facilitate in-depth integration within the region, and promote coordinated regional development, which include the development of a talent hub and a knowledge-innovation hub.

As higher education plays a strategic role in promoting innovation and entrepreneurship (Chow, Hua, and Hung, 2020; Mok, 2022; Tang, 2022), the development of a talent hub and a knowledge-innovation hub in the 'Outline Development Plan for the GBA' is in accordance with the priority for the GBA in the 'China's Education Modernization 2035 Plan', which is to allow for greater mobility of professionals, and facilitate efficient exchange of higher education information and technology within the area.

In 2020, the Ministry of Education (MoE) of China and Guangdong's provincial government released the "Development Plan for Higher Education Cooperation in the Guangdong-Hong Kong-Macau Greater Bay Area", which includes the development of the International Higher Education Demonstration Zone in the GBA.[29] According to the Plan, the GBA will set a good example for comprehensive cooperation in higher education between China mainland, Hong Kong SAR, and Macau SAR. As of August 2023, there are (at least) six cooperation projects that have been set up or are in the pipeline (Table 5).[30]

With over 200 universities and over half a million students, there are some challenges and opportunities associated with the development of the GBA as a talent hub and a knowledge-innovation hub, such as the challenge of promoting higher education, graduate employment, and entrepreneurship in Guangdong province for students who are originally from Hong Kong SAR and Macau SAR (Mok, 2018; Te, 2022; Liu, 2022); the synergies between universities, enterprises and government in promoting innovation-centric entrepreneurship (Mok, Welch, and Kang, 2020); the dynamics of innovation from the 'quadruple helix' of university-industry-government-society relations across borders (Kang and Jiang, 2020); the brain drain, brain gain, and brain circulation for global talent mobility (Mok, Zhang, and Bao, 2022); the potential of a Bay Area University Alliance to enhance the talent-research-industry cooperation mechanism (Mok, 2022); and the unique characteristics of 'one country, two systems', 'three custom territories', and 'three legal systems' (Xie, Liu and McNay, 2023).

[26] See, for example, Douglass (2022).

[27] See, for example, Nelson (2022).

[28] HKSAR government (2017), "Framework Agreement on Deepening Guangdong-Hong Kong-Macau Cooperation in the Development of the Bay Area signed", Press Release. See https://www.info.gov.hk/gia/general/201707/01/P2017070100409.htm

[29] "Greater Bay Area to establish International Higher Education Demonstration Zone", *News Guangdong*, 2020. See https://www.newsgd.com/node_99363c4f3b/f0aec2d229.shtml

[30] Zhou, M. (2021), "Higher education is means to boost region's advancement", *China Daily (HK)*, July. See https://www.chinadailyhk.com/article/226519#Higher-education-is-means-to-boost-region's-advancement



Fig. 6. Greater Bay Area (GBA) cities. *Source*: Greater Bay Area Development Office, Hong Kong SAR.

Table 4
The world's major bay areas: statistical breakdown (2021).

	land area (sq. km)	population (mn)	per-capita GDP (US$)	GDP share of tertiary industry (%)
Guangdong-Hong Kong-Macau Greater Bay Area	56,098	86.62	22,858	64
Tokyo Bay Area	36,898	44.35	46,824	81.4
San Francisco Bay Area	17,887	7.52	161,946	70.8
New York Metropolitan Area	17,312	19.62	96,210	78.6

Source: Hong Kong Trade Development Council (HKTDC).

Table 5
GBA cooperation in higher education.

Hong Kong / Macau university	branch campus in Guangdong Province	year
Chinese University of Hong Kong	Shenzhen	2014
Hong Kong University of Science and Technology	Guangzhou	2022
City University of Hong Kong	Dongguan	2023
Open University of Hong Kong	Zhaoqing	2025
Hong Kong Polytechnic University	Foshan	TBA
Macau University of Science and Technology	Zhuhai	TBA

With the full support of China's Central Government and the three governments in Guangdong Province, Hong Kong SAR, and Macau SAR, the GBA project is expected to be integrated into the overall national development, thereby generating new impetus for growth so as to bring new development opportunities to different sectors of the region. The role of universities, particularly a cluster of successful universities with high-quality research and strong capacity of knowledge transfer, is fundamentally important for the prospects of GBA development.

3.3. The Hainan International Education Innovation Pilot Zone (海南自由贸易港的国际教育创新岛)

Located in the southernmost tip of China, Hainan is an island province, with the Pacific Ocean to the east and the Indian Ocean to the west. Since the early 1980s, special economic zones (SEZs) have become a powerful engine driving economic growth in China. In 1988, Hainan became the 5th and the largest SEZ in China. The evolution of SEZs in China involved the creation of several free trade zones (FTZs). In 2018, Hainan became the 12th FTZ in China. But compared to other coastal SEZs and FTZs, Hainan's economic achievement was moderate, and it remained a less developed province in the coastal area of China.[31]

As education plays an increasingly significant role in regional repositioning in Hainan (Liu and Gao, 2022), the Ministry of Education (MoE) of China and Hainan's provincial government jointly issued the "Circular on Supporting the Implementation Plan of Deepening Education Reform and Opening-up in Hainan" in 2019. According to this circular, China will further open up the sector to overseas education, support Hainan to develop an international education innovation island, and make Hainan a national pilot area for educational innovation and educational development.

In 2020, Chinese authorities released their "Overall Plan for the Construction of Hainan Free Trade Port", which is a large-scale plan to transform the entire island province into a free trade port (FTP) in China. One of the key areas of the Hainan FTP is the Hainan Lingshui Li'an International Education Innovation Pilot Zone, which is an education hub on the Li'an Peninsula in the southeast of Hainan Island. Its main purpose is to create a platform for cooperation between Chinese and foreign universities, provide world-class educational services, and develop innovation in education by attracting quality foreign universities.

Hainan has invested about RMB 40 billion in over 130 cooperation projects and attracted about 25,000 talents in the field of education. Since 2020, the Hainan International Education Innovation Pilot Zone has inked agreements with 22 renowned colleges and universities from home and abroad. As of August 2023, there are eight Sino-foreign joint

[31] An, B. (2019), "Hainan Free Trade Port Development", *ADB Briefs* No. 118, Asian Development Bank (ADB), Manila Philippines, November. See https://www.adb.org/publications/hainan-free-trade-port-development

Table 6
Sino-foreign joint institutes in Hainan.

Sino-foreign joint institute	Chinese university	foreign university	location	year
HNU-ASU Joint International Tourism College	Hainan University (HNU)	Arizona State University (ASU), U.S.	Haikou	2017
UWE College	Hainan Medical University	University of the West of England, UK	Haikou	2022
QTNU-UoN International College	Qiongtai Normal University	University of Northampton, UK	Haikou	2022
Queen Mary School Hainan	Beijing University of Posts and Telecommunications	Queen Mary University of London, UK	Lingshui	2022
Coventry College	Communication University of China	Coventry University, UK	Lingshui	2022
Glasgow College Hainan	University of Electronic Science and Technology of China	University of Glasgow, UK	Lingshui	2022
Alberta International School of Recreation, Sport and Tourism	Beijing Sport University	University of Alberta, Canada	Lingshui	2022
Middlesex College	Minzu University of China	Middlesex University London, UK	Lingshui	2023

Source: China's Ministry of Education (MoE).

institutes in Hainan (Table 6). Among them, three joint institutes are in Haikou, where the capital city of Hainan province is located; and five joint institutes are in Lingshui, where the Hainan International Education Innovation Pilot Zone is located.

Moreover, Hainan has China's first wholly foreign-owned institution: the Hainan Bielefeld University of Applied Sciences, which will be a branch campus of Germany's Hochschule Bielefeld University of Applied Sciences and Arts. It began recruitment of the first batch of students for the autumn semester of 2023,[32] and teaching first takes place in Lingshui, until the branch campus opens in Danzhou in 2025–26. This move is set to foster closer academic and research cooperation between China and Germany, while elevating Hainan's standing as a world-class education hub. [33]

Hainan is the only location in mainland China where foreign universities are not required to partner with local institutions, but can set up fully owned operations. In March 2023, the Ministry of Education (MoE) of China and Hainan's provincial government jointly issued the "Interim Regulation on the Establishment of Institutes by Overseas Higher Education Institutions (HEIs) in the Hainan Free Trade Port", [34] which allows and regulates HEIs outside mainland China to establish and operate universities independently within the Hainan FTP. This is an important step in opening up China's higher education system.

The education reform and education innovation in the Hainan FTP is a work-in-progress. Challenges and opportunities could co-exist, including the challenge of 'Internationalization at Home' (Qu and Dai, 2024) in running Sino-foreign joint institutes, the challenge of glocalization in running wholly foreign-owned institutions; and the opportunity of further development of the vocational education sector to better suit the needs of the local industries in Hainan. With the full support of the Chinese government, Hainan has the potential to transform from a student hub into a talent hub in the future.

4. Conclusion

Internationalization is changing the world of higher education, and globalization is changing the world of internationalization (Knight, 2004). Both as an actor and as a reactor to globalization, internationalization has not only transformed higher education, but it has undergone fundamental changes itself. There are two pillars in the internationalization process of higher education, which include 'Internationalization at Home' (IaH, Nilsson, 2003; Qu and Dai, 2024) and 'crossborder education'. These two pillars are closely linked and interdependent. Within the context of crossborder education, there are three generations of development, which include academic mobility, program and provider mobility, and education hubs. Building on and extending the first and second generations of crossborder education activities, education hubs constitute the *third* generation of crossborder education initiatives, which demonstrate the changing and responsive nature of internationalization to the complex realities of our more globalized world. Based on the rationales of the activities, there are three different but linked models of education hubs, which include student hub, talent hub, and knowledge-innovation hub (Knight, 2011).

As a new phenomenon in the landscape of higher education, education hubs are appearing in different locations around the world, each with its own individual political, economic, and cultural context. In the literature on education hubs, most studies have focused on Southeast Asia and Middle East (Table 1), and relatively few studies have examined the case of China, which is both a source and a hub for international students (Wei, 2013; Ma, 2017; Wen, Hu, and Hao, 2018; Wen and Hu, 2019; Tian and Liu, 2021). To address this gap in the literature, we explore the emergence of China as an international education hub, especially with respect to its impact on development. Against the backdrop of China's national strategy to develop the country's international competitiveness in education, we conduct case studies of three education hubs in China, which include (i) the International Collaborative Education Exemplary Zone by the International Campus of Zhejiang University (iZJU) in the Yangtze River Delta; (ii) the International Higher Education Demonstration Zone in the Guangdong-Hong Kong--Macau Greater Bay Area; and (iii) the International Education Innovation Pilot Zone in the Hainan Free Trade Port (Table 7).

Our first case study is the International Collaborative Education Exemplary Zone led by the International Campus of Zhejiang University (iZJU) in the Yangtze River Delta. It is a student hub, with the feature of 'Internationalization at Home' for Chinese students (Qu and Dai, 2024), and the brand of 'Study in Haining' for international students. Moreover, it aims to become a knowledge-innovation hub and a talent hub, by cultivating a knowledge, innovation, and entrepreneurial ecosystem with the Juanhu International Science Park that is conducive to attracting international talent in education and science. The regional development model of iZJU is like a single-core-driven model with iZJU as the core, which leverages the strength of ZJU's network model of campuses and iZJU's partnership model with world-class universities (Fig. 4), unleashes the region's technological and innovational capacity, and integrates the development of iZJU and the Haining city. It sets a role model for the transformation of other county-level cities in China.

Our second case study is the International Higher Education Demonstration Zone in the Guangdong-Hong Kong-Macau Greater Bay Area (GBA). As higher education plays a strategic role in promoting innovation and entrepreneurship in the GBA (Chow, Hua, and Hung, 2020; Mok, 2022; Tang, 2022), the objectives for the GBA are to be a talent hub that allows for greater mobility of professionals, and a knowledge-innovation hub that facilitates efficient exchange of higher education information and technology within the area. With over 200

[32] "China's 1st foreign-run college to be built in Hainan", eHainan, 2022. See http://www.ehainan.gov.cn/2022–12/16/c_840272.htm

[33] "Hainan releases plan to attract foreign universities", TropicalHainan, 2023. See https://www.tropicalhainan.com/hainan-releases-plan-to-attract-foreign-universities/

[34] "New regulation issued on international branch campuses in Hainan", Ministry of Education (MoE), Beijing, April 2023. See http://en.moe.gov.cn/news/press_releases/202304/t20230427_1057585.html

Table 7
Case Studies of Education Hubs in China – summary.

education hub	type
The iZJU International Collaborative Education Exemplary Zone	student hub, knowledge-innovation hub, talent hub
The GBA International Higher Education Demonstration Zone	talent hub, knowledge-innovation hub
The Hainan International Education Innovation Pilot Zone	student hub -> talent hub

universities and over half a million students, the development of the GBA International Higher Education Demonstration Zone will set a good example for comprehensive cooperation in higher education between China mainland, Hong Kong SAR, and Macau SAR, with the unique characteristics of 'one country, two systems', 'three custom territories', and 'three legal systems' (Xie, Liu and McNay, 2023).

Our third case study is the Hainan International Education Innovation Pilot Zone in the Hainan Free Trade Port (FTP). It is a student hub with the main purpose to create a platform for cooperation between Chinese and foreign universities, provide world-class educational services, and develop innovation in education by attracting quality foreign universities. As of August 2023, there are five Sino-foreign joint institutes in the Hainan International Education Innovation Pilot Zone. China's first wholly foreign-owned institution will also be in the Hainan FTP. As education plays an increasingly significant role in regional repositioning in Hainan (Liu and Gao, 2022), the regional development model of the Hainan International Education Innovation Pilot Zone and the Hainan FTP is to experiment with education opening up and innovation so as to promote high-quality development and growth in Hainan.

The global advancement of the knowledge economy (Altbach, 2013) and the growing internationalization of higher education has resulted in the emergence of a number of education hubs in different regions of the world. Our case studies of three education hubs in China show that these hubs face similar challenges and opportunities. For a student hub, a common challenge is how to develop an ecosystem that is conducive to attracting reputable international education providers; and a common opportunity is posed by the rising global demand for quality international education nearer to home, that is, 'Internationalization at Home' (Qu and Dai, 2024). For a talent hub, a common challenge and opportunity is the brain drain and brain gain for global talent mobility (Mok, Zhang, and Bao, 2022). For a knowledge-innovation hub, a common opportunity is the dynamics of innovation from the 'triple helix' model of university-industry-government relations (Etzkowitz and Leydesdorff, 2000), and the 'quadruple helix' of university-industry-government-society relations (Kang and Jiang, 2020), which is especially important for countries keen to shift to a knowledge and service-based economy, and gain a competitive edge in the region and beyond.

In the aftermath of the COVID-19 global pandemic, the advancement in digital technology and innovation has driven a 'virtual' internationalization of higher education, with universities expanding their digital footprints overseas, accelerating their distance education offerings, exploring virtual exchange programs, and adopting virtual communication platforms (Wernick, 2023). As the three case studies in our paper are fledging education hubs from China, future studies could be conducted to further examine how they evolve and thrive in response to an uncertain and dynamic global environment.

Authors' Statement

The authors have no competing interests to declare.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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